

Physics Mock Test

Moving Charges and Magnetism

Class 12 | UNIT 3

Marks: 24

Time: 60 Minutes

Coverage: Magnetic force, Biot-Savart Law, Ampere's Circuital Law, Solenoids, Cyclotron, and Force on Currents.

Show your calculation steps clearly to ensure maximum marks in Section 2.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Magnetic Force Optimization

A charged particle moving in a magnetic field experiences a force $F = q(v \times B)$. This force magnitude is maximum when:

- a) Velocity v is parallel to B
- b) Velocity v is perpendicular to B
- c) Velocity v is at 45° to B
- d) Velocity v is zero

2. Field Determination

The direction of the magnetic field vector produced by a straight current-carrying wire is found using:

- a) Right-hand thumb rule
- b) Fleming's left-hand rule
- c) Fleming's right-hand rule
- d) Lenz's Law

3. Biot-Savart Law Foundation

Biot-Savart law mathematically describes the magnetic field produced by:

- a) A point charge
- b) An infinitesimal current element
- c) A permanent bar magnet
- d) A stationary dipole

4. Ampere's Circuital Law

According to Ampere's law, the line integral of magnetic field $B \cdot dl$ around any closed loop is proportional to:

- a) $\mu_0 \times$ Net Current enclosed
- b) Net Current / μ_0
- c) $\mu_0 \epsilon_0 \times$ Current
- d) Magnetic Flux enclosed

5. Force Scaling

The force on a current wire is $F = I L B \sin\theta$. If the current I is doubled and the magnetic field B is halved, the magnetic force:

- a) Remains the same
- b) Is doubled
- c) Is halved
- d) Becomes four times

6. Cyclotron Motion

In a cyclotron, under the influence of a uniform magnetic field, the charged particles describe:

- a) Parabolic paths
- b) Circular paths
- c) Elliptical paths
- d) Constant linear motion

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Lorentz Force for a Proton

A proton (charge $+1.6 \times 10^{-19}$ C) moves with a velocity of 10^6 m/s perpendicular to a magnetic field of 0.1 T. Calculate the magnitude of the force acting on the proton.

8. Field from an Infinite Wire

A long straight wire carries a current of 5 A. Calculate the strength of the magnetic field at a point located 2 cm from the axis of the wire.

($\mu_0 = 4\pi \times 10^{-7}$ T•m/A)

9. Circular Loop Magnetics

Calculate the magnetic field at the center of a circular loop of radius 10 cm that carries a current of 2 A.

10. Internal Field of a Solenoid

A solenoid has a winding density of 500 turns per meter and carries a current of 2 A. Determine the magnetic field strength deep inside the solenoid.

11. Mechanical Force on a Wire

A straight wire of length 20 cm carrying a current of 3 A is placed perpendicular to a magnetic field of 0.5 T. Calculate the force exerted on the wire.

12. Cyclotron Frequency

Protons are accelerated in a cyclotron where the magnetic field is 1 T. Find the cyclotron frequency.

($m_p = 1.67 \times 10^{-27}$ kg, $q = 1.6 \times 10^{-19}$ C)

Answer Key & Solutions

1. b) $\mathbf{v} \perp \mathbf{B}$

$F = qvB \sin\theta$. Maximum when $\sin\theta = 1$ ($\theta = 90^\circ$).

2. a) Right-hand thumb rule

Thumb points to current, fingers curl in direction of $\mathbf{B}$.

3. b) Infinitesimal current element

$dB = (\mu_0/4\pi) (I dl \sin\theta / r^2)$.

4. a) $\mu_0 \times I_{\text{enclosed}}$

$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$.

5. a) Remains same

$F' = (2I) L (B/2) \sin\theta = BIL \sin\theta = F$.

6. b) Circular paths

Magnetic force provides the necessary centripetal force.

7. $1.6 \times 10^{-14} \text{ N}$

$F = qvB = (1.6 \times 10^{-19})(10^6)(0.1) = 1.6 \times 10^{-14} \text{ N}$.

8. $5 \times 10^{-5} \text{ T}$

$B = \mu_0 I / 2\pi r = (4\pi \times 10^{-7} \times 5) / (2\pi \times 0.02) = (10 \times 10^{-7}) / 0.02 = 5 \times 10^{-5} \text{ T}$.

9. $4\pi \times 10^{-6} \text{ T}$ ($\approx 1.26 \times 10^{-5} \text{ T}$)

$B = \mu_0 I / 2R = (4\pi \times 10^{-7} \times 2) / (2 \times 0.1) = 4\pi \times 10^{-6} \text{ T}$.

10. $4\pi \times 10^{-4} \text{ T}$ ($\approx 1.257 \times 10^{-3} \text{ T}$)

$B = \mu_0 nI = (4\pi \times 10^{-7})(500)(2) = 4\pi \times 10^{-4} \text{ T}$.

11. 0.3 N

$F = BIL = (0.5)(3)(0.2) = 0.3 \text{ N}$.

12. $1.52 \times 10^7 \text{ Hz}$ (15.2 MHz)

$f = qB / (2\pi m) = (1.6 \times 10^{-19} \times 1) / (2 \times 3.14 \times 1.67 \times 10^{-27}) \approx 1.52 \times 10^7 \text{ Hz}$.